

Ariun-Erdene Tumurchuluun, CV

I am an MSc graduate and AI engineer with a research focus on alignment and fairness challenges in state-of-the-art LLMs. Specifically, my main interest is understanding how state-of-the-art multilingual models internally represent alignment and fairness-related concepts and whether these representations are universal, language-specific, or both, and how mechanistic interpretability tools can help us understand and mitigate the underlying issues. My path into this area began in software engineering, where I was fortunate enough to meet my undergraduate supervisor, Khuyagbaatar, who encouraged me to pursue NLP more deeply and supported my application to the prestigious Language and Communication Technologies program, which is a double master's degree that became the foundation of my research journey. In my second year, a software project course taught by DFKI researchers introduced me to mechanistic interpretability, where I explored key tools and read Anthropic's work on scaling monosemanticity, which led me to propose a thesis grounded in this framework. My thesis supervisors welcomed the idea, and together with my daily supervisor Koel, we carried out a project which we eventually submitted to the MRL workshop at EMNLP 2025 in Suzhou, China, where it received the Best Paper Award. Since then, we have stayed in touch, and we are currently investigating stereotype bias, spanning gender, age, sexual orientation, physical appearance, and more, across multiple model families and languages, asking to what extent these biases are represented universally versus language-dependently, and whether the patterns hold consistently across different architectures. This work is currently under review at CoLM 2026. I am now seeking a PhD position where I can extend this line of exploration into projects that are meaningful for both academia and industry, ideally within a research environment with strong expertise in mechanistic interpretability, AI fairness, alignment, and safety.

CONTACT

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Location Ulaanbaatar, Mongolia

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RESEARCH INTERESTS

AI fairness, alignment, and safety; multilingual LLMs; mechanistic interpretability; representation analysis; sparse autoencoders and monosemantic features; causal tracing; attribution patching; path patching; probing; model steering.

EDUCATION

Oct 2023 – Sep 2025 **M.Sc., Language Science and Technology**, *Universität des Saarlandes*, Saarbrücken, Germany.

Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.

GPA: 2.0/5.0 (German scale; 1.0 = best).

Coursework: Multimodal Language Understanding, Mechanistic Interpretability, Speech Science, Software Project, Computational Linguistics.

- Oct 2023 – Sep 2025 **M.Sc., Language Technologies and Computational Linguistics**, *Charles University*, Prague, Czech Republic.
 Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.
 GPA: 1.91/4.0 (Czech scale; 1.0 = best), Summa Cum Laude.
 Coursework: Deep Learning, Unsupervised Machine Learning, Speech Recognition and Generation, Statistical Methods in NLP.
- Sep 2018 – May 2022 **B.Eng., Software Engineering**, *National University of Mongolia*, Ulaanbaatar, Mongolia.
 GPA: 3.8/4.0 (top 1%), Summa Cum Laude.
 Coursework: Web Development, Object-Oriented Programming, Data Structures, Algorithms, Databases, Software Engineering, Project Management, Agile Development, Cloud Computing, Cybersecurity.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed

1. **Ariun-Erdene Tumurchuluun**, Yusser Al Ghussin, David Mareček, Josef Van Genabith, and Koel Dutta Chowdhury. 2025. TenseLoC: Tense Localization and Control in a Multilingual LLM. In *Proceedings of the 5th Workshop on Multilingual Representation Learning*, pages 243–264, Suzhou, China. Association for Computational Linguistics.

Manuscripts under review

1. **Ariun-Erdene Tumurchuluun***, Yusser Al Ghussin*, Pinzhen Chen, Josef Van Genabith, and Koel Dutta Chowdhury. Under review at CoLM 2026, San Francisco, USA (* equal contribution).

SELECTED RESEARCH PROJECTS

- Nov 2025 – present **There Is No Universal Language for Bias but a Common Fix.**
 Investigated whether stereotype bias in state-of-the-art LLMs is shared across languages or language-specific across model families. Designed analyses spanning gender, age, sexual orientation, physical appearance, and related attributes; manuscript under review at CoLM 2026.
- Jun 2025 – Oct 2025 **TenseLoC: tense localization and control in a multilingual LLM.**
 Studied internal representations of grammatical tense (morphology, syntax, semantics) in Llama 3.1 across eight languages using probing, causal tracing, sparse autoencoders, and activation steering grounded in identified tense features. Outcome: peer-reviewed publication at MRL @ EMNLP 2025 (Best Paper Award).
- Apr 2025 **Word-order encoding in a multilingual BERT.**
 Implemented diagnostic classifiers to quantify mBERT sensitivity to word-order perturbations in six typologically diverse languages.
- Mar 2022 **Myers-Briggs types from handwriting.**
 Collected handwriting and self-report personality data (~50 participants), preprocessed character images, and trained a supervised classifier (~61% test accuracy).

INDUSTRY EXPERIENCE

- Nov 2025 – present **AI Engineer**, *Egune AI LLC*, Ulaanbaatar, Mongolia.
Built NLP systems in Mongolian for spelling and grammar correction, POS tagging, word sense and homonym disambiguation, translation evaluation, RAG-based document retrieval, and text rewriting. Curated and generated training data; fine-tuned models with full fine-tuning, LoRA, and QLoRA (Axolotl); qualitative and quantitative evaluation; inference with Hugging Face Transformers and vLLM; APIs and demo documentations; used Playwright for data collection.
- Jun 2021 – Jul 2023 **Software Engineer**, *AI Lab LLC*, Ulaanbaatar, Mongolia.
Full-stack engineer for a financial clients, where I delivered and maintained three production web applications (investment and funding); implemented microservice features, search improvements, and OAuth SSO (e.g., Firebase Authentication, Google Cloud Functions); migrated identity from Keycloak to Firebase Auth; maintained a Django crowdfunding platform. Stack: Python, PHP, PostgreSQL, AWS, Firebase, Redis, HTML/CSS/JavaScript, automated unit testing.

FUNDING AND SCHOLARSHIPS

- 2023 – 2025 **Erasmus Mundus Scholarship**.
Joint master's scholarship in Language & Communication Technologies; total support approximately EUR 55,000 (stipend, tuition, transportation, insurance, and related fees and allowances).

AWARDS AND HONORS

- Best Paper Award, 5th Workshop on Multilingual Representation Learning @ EMNLP, 2025
- Most Advanced Graduate Award, National University of Mongolia, 2022

TECHNICAL SKILLS

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| ML & NLP | Python, PyTorch, Hugging Face Transformers, vLLM, fine-tuning (full, LoRA, QLoRA) with Axolotl and ms-swift, evaluation and error analysis, data curation, RAG pipelines, mechanistic tools (e.g., probing, causal tracing, sparse autoencoders, steering). |
| Engineering | Git, Linux, GPU clusters, visualization, unit and integration testing, REST APIs, PostgreSQL, Redis, Django, Vue.js, cloud backends (AWS, Firebase, Google Cloud Functions), Playwright. |
| Languages | English (professional); Mongolian (native); German (beginner); French (beginner); Russian (beginner); Japanese (beginner). |
| References | Available upon request. |